Comment on the solutions to the Poisson equation

The potential from the homogeneously charged sphere, as well as from the homogeneously charged shell are continues over the whole interval where we looked at them, and so are their first derivatives. The second derivatives on the other hand are discontinuous at the places where the charge distribution is discontinuous (obviously).

For example, for the potential from the sphere we have in the inner region, $r < R$, the analytical solution

$$V(r) = \frac{Q}{4\pi\varepsilon_0 R}\left(\frac{3}{2} - \frac{r^2}{2R^2}\right) \tag{1}$$

and in the outer region

$$V(r) = \frac{Q}{4\pi\varepsilon_0 r}. \tag{2}$$

These expressions coincide when $r = R$. Looking at the derivative, we find that in the inner region it is:

$$\frac{dV}{dr} = -\frac{Q}{4\pi\varepsilon_0}\frac{r}{R^3} \tag{3}$$

and in the outer region

$$\frac{dV}{dr} = -\frac{Q}{4\pi\varepsilon_0 r^2}, \tag{4}$$

thus also these coincide at $r = R$. The second order derivative on the other hand is

$$\frac{d^2V}{dr^2} = -\frac{Q}{4\pi\varepsilon_0}\frac{1}{R^3} \tag{5}$$

in the inner regions and

$$\frac{d^2V}{dr^2} = 2\frac{Q}{4\pi\varepsilon_0 r^3}, \tag{6}$$

in the outer region and do not coincide.

When we use B-splines with $k = 4$ these meets the requirement in all points, except these points where the charge distribution is discontinuous. This, because we expand the potential in functions that have a continues second order derivative, even where it should not be continues. If you calculate the second order derivative of your solution ϕ, ($d^2\phi/dr^2$ with $\phi = rV$), also in the points between the knot points you will find a charge distribution that is much smoother than the one you set out to solve for.

How can we improve the situation? One possibility is to use two knot points in the physical points where the charge distribution is discontinuous and then ensure that the left and right second order derivative is different. Another - in some sense simpler - solution is just to put some knot points really close to each other at the boundaries of the charged sphere/shell. In the picture below the red curve is obtained with one knot point at $R_{inner} = 5$ and then one extra point at $R_{\text{inner}} - \Delta$ and one at $R_{\text{inner}} + \Delta$, and the same for R_{outer}. In the example $\Delta = 10^{-6}$. The results is then spot on the analytical solution with just five other knot points (uniformly distributed) in each region (inside, within or outside the shell). The constant potential in the inner region coincide with the analytical solution to seven digits .

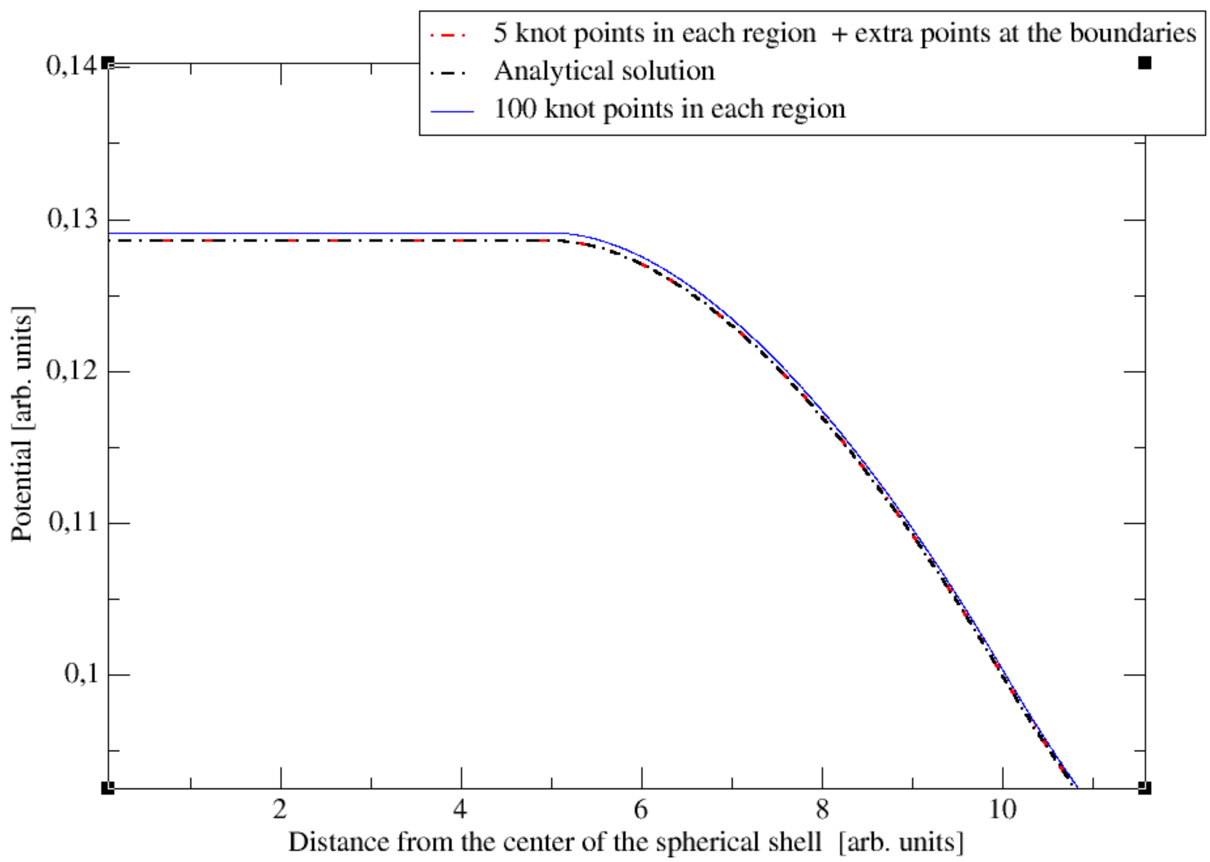

Figure 1: Potential from a homogeneously charged shell with $R_{\mathrm{inner}} = 5$ and $R_{\mathrm{outer}} = 10$.